Study: Numerical modelling of Wave-vegetation interactions

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Abstract

This project is a pre study of wave-vegetation interactions. The aim is to evaluate the impact of vegetation on the of the sea on the waves behavior, like the change of wave height, period, velocity etc...

This project is also here to support my candidature to the PhD Scolarship on wave-vegetation interactions at DTU. The aim is not to support directly a strong and valid model, but to demonstrate my abilities to choose models, handle complex equations and develop numerical modelling on this type of topic.

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1 Introduction

1.1 Objectives

The objective of this project is to investigate the influence of submerged coastal vegetation on wave propagation and attenuation through numerical modelling.

The study begins with a simplified wave-energy model based on linear wave theory and Morison's equation, representing vegetation as an array of rigid vertical cylinders. The aim is to predict wave-height attenuation and investigate the influence of vegetation properties and wave conditions.

2 First Model: Wave Attenuation through Rigid Cylinders under Regular Waves

I first developed a simplified representation of wave-vegetation interaction.

The vegetation is represented as an array of rigid vertical cylinders. Wave attenuation is calculated from the energy dissipated by hydrodynamic drag, using Morison's equation and the wave energy balance.

The model predicts the evolution of wave height $H(x)$ along the vegetated area.

2.1 Main Assumptions

Wave assumptions:

- Regular, progressive waves propagating along the x-axis.
- Linear wave theory (Airy).
- Constant water depth $h = 5$ m.
- Constant wave period $T = 4.0$ s.
- Wave breaking and reflection are neglected.

Vegetation assumptions:

- Vegetation is represented as rigid, fixed, vertical cylinders.
- Vegetation is present between $x = 0$m and $x = 50$m.
- All cylinders have the same diameter $d = 0.1$ m and height $h_v = 1$ m.
- Constant drag coefficient $C_D = 1.2$ and inertia coefficient $C_M = 1.0$.
- Cylinders are uniformly distributed, with a density $n = 100$ $(1/\mathrm{m}^2)$.
- Interactions between neighbouring cylinders are not explicitly resolved.

2.2 Linear Wave Theory

The wave number is obtained from the linear dispersion relation:

$$\omega^2 = gk \tanh(kh) \tag{2.1}$$

where:

- $\omega = 2\pi/T$ is the angular wave frequency (rad/s).
- g is the gravitational acceleration $(\mathrm{m/s}^2)$.

- $k = 2\pi/L$ is the wave number (rad/m).
- h is the water depth (m).
- L is the wavelength (m).

The horizontal orbital velocity is:

$$u(x,z,t) = \frac{H(x)}{2}\omega\frac{\cosh(k(z+h))}{\sinh(kh)}\cos(kx-\omega t) \tag{2.2}$$

The horizontal orbital acceleration is:

$$\frac{\partial u}{\partial t}(x,z,t) = -\frac{H(x)}{2}\omega^2\frac{\cosh(k(z+h))}{\sinh(kh)}\sin(kx-\omega t) \tag{2.3}$$

where:

- $H(x)$ is the local wave height (m).
- x is the horizontal position along the wave propagation direction (m).
- z is the vertical coordinate, with $z = 0$ at the still water level (m).
- t is time (s).
- u is the horizontal orbital velocity (m/s).
- $\partial u/\partial t$ is the local horizontal acceleration (m/s^2).

2.3 Morison's Equation and Fluid Force on the Vegetation

Morison's equation is used to calculate the horizontal hydrodynamic force exerted by the fluid on a rigid vertical cylinder.

The force per unit length of cylinder is:

$$f_M = \underbrace{\frac{1}{2}\rho_w C_D d|u|u}_{\text{Drag force}} + \underbrace{\rho_w C_M \frac{\pi d^2}{4}\frac{\partial u}{\partial t}}_{\text{Inertia force}} \tag{2.4}$$

where:

- ρ_w is the water density (kg/m^3).
- C_D is the drag coefficient (-).
- C_M is the inertia coefficient (-).
- d is the cylinder diameter (m).
- u is the horizontal fluid velocity (m/s).

- $\partial u/\partial t$ is the horizontal fluid acceleration (m/s^2).
- f_M is the hydrodynamic force per unit cylinder length (N/m).

Morison's equation gives the force applied by the fluid on one cylinder per unit length. We can then obtain the force applied to the vegetated region per unit volume by multiplying by the cylinder density n:

$$f_M^{vol} = \frac{1}{2}\rho_w C_D n d|u|u + \rho_w C_M n \frac{\pi d^2}{4}\frac{\partial u}{\partial t} \tag{2.5}$$

where n is the number of cylinders per unit horizontal bed area (stems/m^2), and f_M^{vol} is expressed in N/m^3.

Using Newton's third law, the force applied by the vegetation on the fluid is equal and opposite to the force applied by the fluid on the vegetation:

$$f_{\text{veg}\to\text{fluid}} = -f_M^{vol} \tag{2.6}$$

2.4 Wave Energy Dissipation

The mechanical power extracted from the fluid per unit volume by the vegetation is calculated from the hydrodynamic force and the fluid velocity:

$$p_M = f_M^{vol} u \tag{2.7}$$

where:

- p_M is the instantaneous mechanical power extracted from the fluid per unit volume (W/m^3).
- f_M^{vol} is the force exerted by the fluid on the vegetation per unit volume (N/m^3).
- u is the horizontal fluid velocity (m/s).

The mean power extracted over one wave period is given by:

$$\overline{p_M}(x,z) = \frac{1}{T}\int_0^T f_M^{vol}(x,z,t)\, u(x,z,t)\, dt \tag{2.8}$$

The force f_M^{vol} is composed of two terms: drag and inertia. The mean contribution of the inertia term is zero for a periodic velocity field and a constant inertia coefficient, since:

$$\overline{u\frac{\partial u}{\partial t}} = \frac{1}{T}\int_0^T u\frac{\partial u}{\partial t}\, dt = \frac{1}{2T}\left[u^2\right]_0^T = 0 \tag{2.9}$$

Therefore, the mean power dissipated by the vegetation comes only from the drag contribution and is equal to:

$$\overline{p_D}(x,z) = \frac{1}{2}\rho_w C_D n d\overline{|u(x,z,t)|^3} \tag{2.10}$$

By integrating over the vegetation height, we obtain the vegetation-induced dissipation per unit horizontal area:

$$D_v(x) = \frac{1}{T} \int_{-h}^{-h+h_v} \int_0^T \frac{1}{2} \rho_w C_D n d |u(x,z,t)|^3 \, dt \, dz \tag{2.11}$$

where $D_v(x)$ is the mean wave energy dissipation rate per unit horizontal area ($\mathrm{W/m^2}$).

2.5 Wave Energy Attenuation

To compute wave attenuation, we consider the energy flux over a small horizontal element dx and account for the energy dissipated by the vegetation.

From linear wave theory, the mean wave energy per unit horizontal area is:

$$E(x) = \frac{1}{8} \rho_w g H(x)^2 \tag{2.12}$$

where $E(x)$ is expressed in $\mathrm{J/m^2}$.

The wave energy flux per unit width is defined as:

$$F(x) = E(x) c_g \tag{2.13}$$

where c_g is the group velocity, given by:

$$c_g = \frac{\omega}{2k} \left(1 + \frac{2kh}{\sinh(2kh)} \right) \tag{2.14}$$

The energy evolution along the horizontal direction is then obtained from the energy balance:

$$F(x) - F(x + dx) = D_v(x) \, dx \tag{2.15}$$

Taking the limit as dx approaches zero gives:

$$\frac{d(E c_g)}{dx} = -D_v(x) \tag{2.16}$$

Since the water depth and wave period are constant, the group velocity c_g is also constant. Substituting Eq. (2.12) gives:

$$\frac{1}{8} \rho_w g c_g \frac{d(H^2)}{dx} = -D_v(x) \tag{2.17}$$

This finally gives the differential equation governing the evolution of the wave height:

$$\boxed{\frac{dH}{dx} = -\frac{4 D_v(x, H)}{\rho_w g c_g H(x)}} \tag{2.18}$$

The equation is solved numerically using the `solve_ivp` function from `scipy.integrate`, with the initial condition:

$$H(0) = H_0 \tag{2.19}$$

At each integration step, the local wave height is used to update the orbital velocities and the corresponding energy dissipation. The numerical solution provides the evolution of wave height $H(x)$ along the vegetated area.

2.6 Results

The evolution of the wave height is as follow:

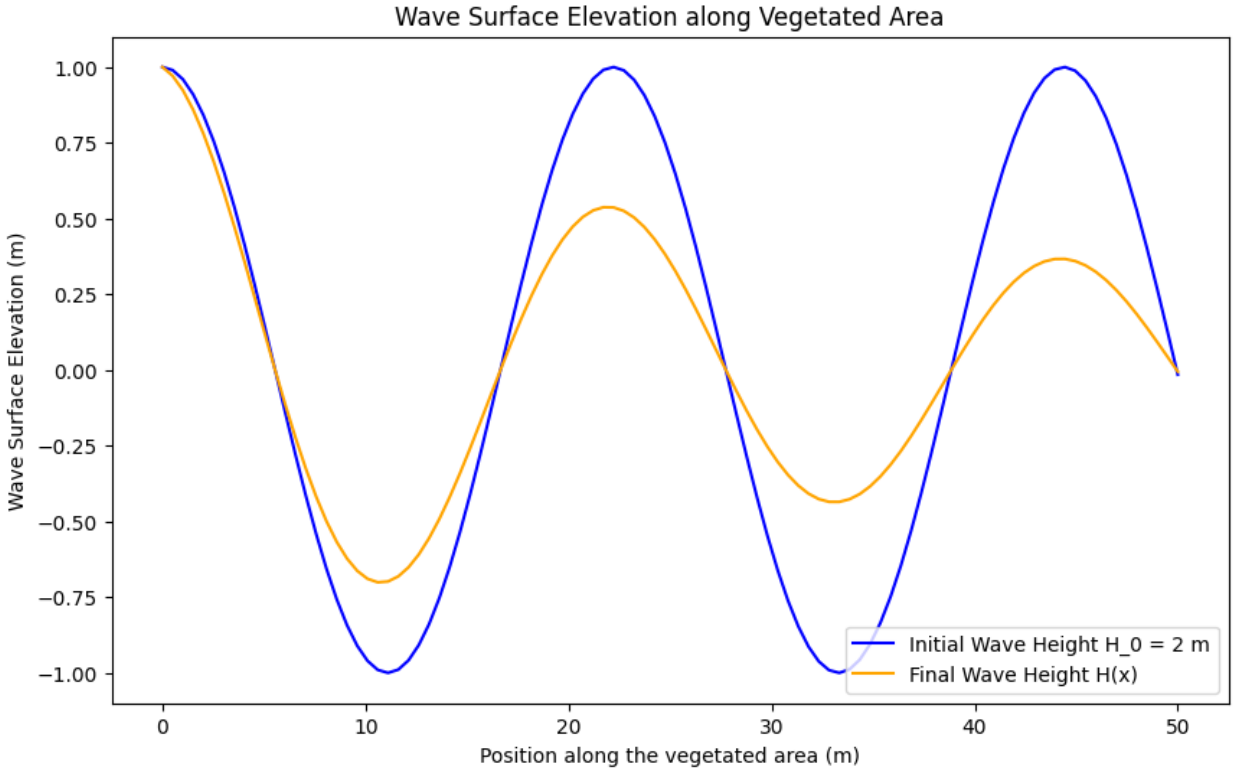


Figure 1: Wave Height evolution.